

Acknowledgements

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Chapter 1

Introduction

1.1 Motivation

Financial time series data is never fully observed. Limit-order book feeds drop quotes during bursts of order cancellations. Equity markets suspend trading when prices move too violently. Over-the-counter bond markets leave entire yield-curve tenors unquoted for days. Implied volatility surfaces contain large regions where no liquid option trade has occurred. Macro and alternative data arrive at mixed, irregular frequencies, with publication lags that create structural gaps at the daily level. This missing data is pervasive and often elusive, and gets in the way of advanced analysis. Thus practitioners or a researchers face the same decision: what value should be attributed to the missing observation, and how much uncertainty should be attached to that attribution?

The simplest and most common solution is to carry forward the last observed value, but this can lead to systematic mispricing, underestimation of risk, and distortion of correlations among assets.

We aim to take a different path, one that looks for solutions that aim to preserve as much as possible the statistical properties of the series, in order to maintain the integrity of the underlying data and consequently the robustness of any downstream model that relies on it. Since this can be considered an inference problem, we will also explore ML algorithms, and in classical machine learning approach, we will focus on quality of inference rather than interpretability of the model.

In this thesis we will test these different methods of imputation, and evaluate them not only on the accuracy of the imputation itself (MSE,...), but also on the quality of the downstream model that uses the imputed data, and how far it deviates from the case where all data is observed.

1.2 Literature review

1.3 Research questions

1.4 Contributions

1.5 Outline of the thesis

Chapter 2

Methodology

As already mentioned, missing data can arise in countless different ways, and the choice of methodology depends on the nature of the missingness, the type of data, and the specific goals of the analysis. We will focus on a subset of cases, namely those that involve imputation of the missing values with "information" from the whole dataset (no causal constraint), with in mind the goal of training a downstream model that is as close as possible to the one that would have been trained on the complete dataset. Our choice of data, models, and evaluation methods will be guided by these constraints. We will nevertheless mention when/how models should be tweaked to solve adjacent problems.

2.1 Data

As our main baseline, we chose the most studied and reliable dataset in finance: the end-of-day prices of the S&P500, in this case fetched with `yfinance` from 1/1/2010 to 30/5/2026 (potrei cambiare le date). It clearly is not a dataset inflicted with the curse of missing data; the only structural ones being occasional circuit breaks and listings/delistings from the index. To obtain a full dataset with no "real" missing data, we removed each stock that contained at least one missing observation. Furthermore, we selected a sample of 50 stocks from the index, stratified across industries; this is a sensible choice because:

1. with more 400 stocks the covariance matrix Σ quickly becomes unreliable (noisy covariances, skewed eigenvalues, unstable inversion. . .) as the number of observations shrinks;
2. it is easier to check whether the ground data is actually correct;
3. model training requires considerably less memory, in a time where RAM and GPU prices are skyrocketing;
4. it is well known that a well constructed portfolio with more than 30 stocks is already sufficiently diversified (since volatility scales with the inverse of the square root of

the number of securities).

This dataset will serve as the ground truth against which we will compare our results. In parallel, we will modify it by inserting fake missing observations according to different rules, as we define in the subsequent sections. Now an important choice is whether to operate in the prices "space" or the returns space. There are positives and negatives about each. For one, the raw dataset is prices, and in practice missing data will come in the form of missing prices.

2.2 Models

2.3 Evaluation Methods

Chapter 3

Imputation Methods

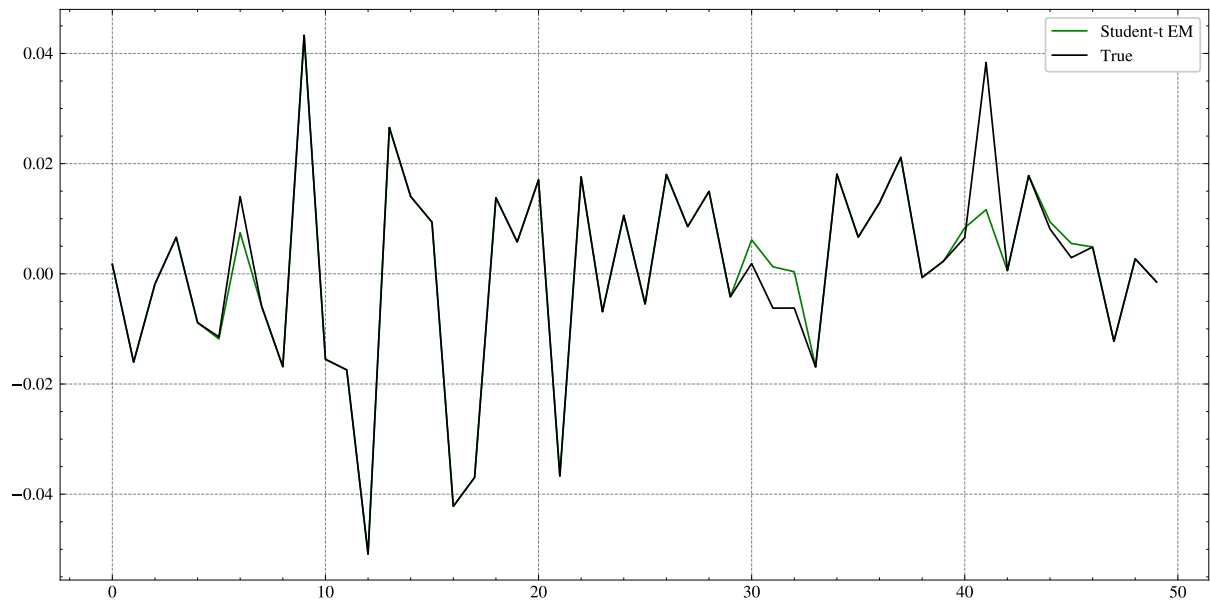


Figure 3.1: Examples of Imputed Data

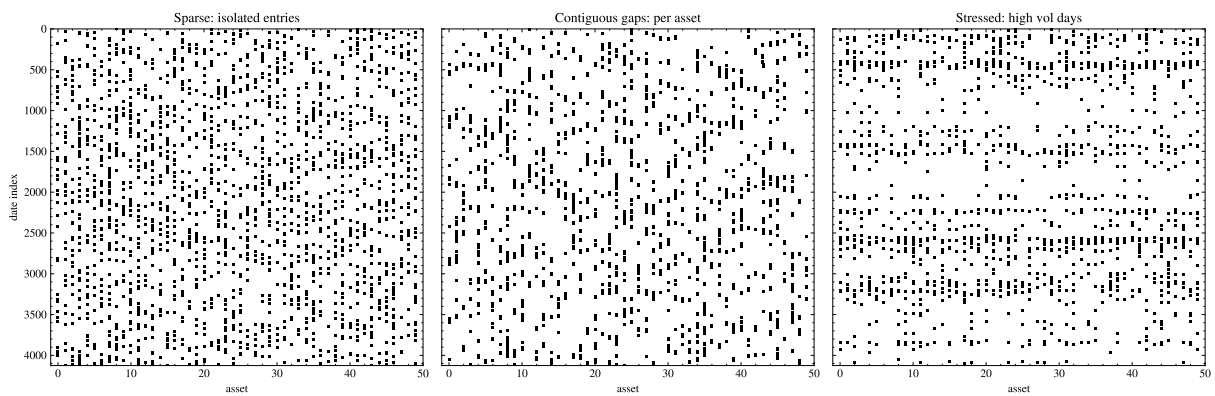


Figure 3.2: Missingness Examples

Chapter 4

Results and Discussion

Chapter 5

Conclusion

5.1 Summary

5.2 Limitations

5.3 Future Work

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